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PSAC Exam Practice · Predicted Specimen Paper

2026

SCIENCE

Grade 5 · Paper 2

Plant Parts and Functions | Properties of Materials | Sound and Vibration

Time Allowed: **1 hour**

Total Marks: **50**

Name: _____

Class: _____ Date: _____

INSTRUCTIONS TO CANDIDATES

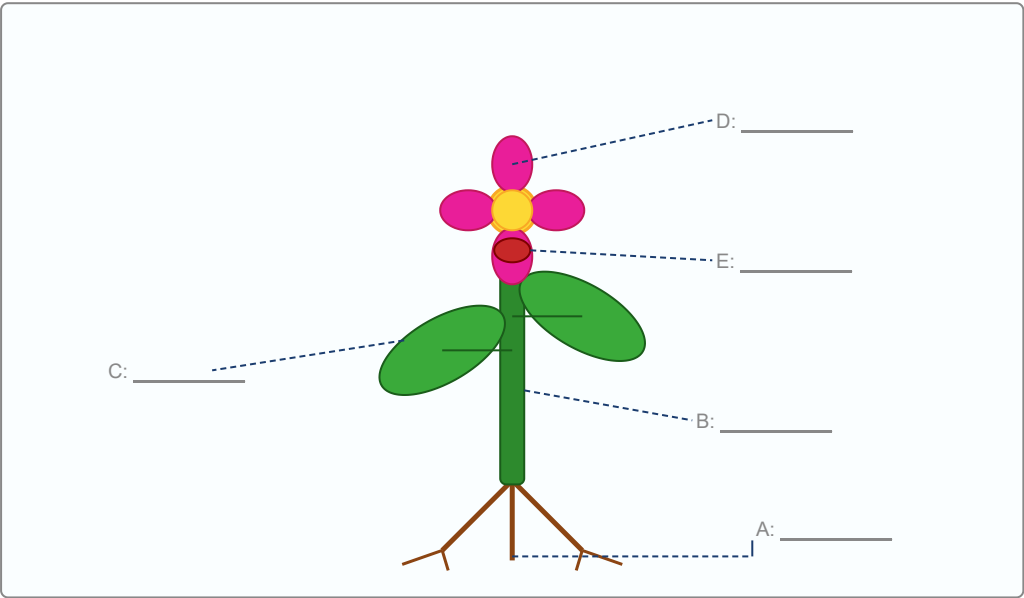
1. Write your name, class and date in the spaces provided above.
2. Answer **ALL** questions in **all three sections**.
3. Write your answers clearly in the spaces provided.
4. The mark for each question or part question is shown in square brackets [].
5. Diagrams should be clear and neatly labelled.

Section	Topic	Available	Score
A	Living Things — Plant Parts and Functions	20	
B	Matter and Materials — Properties of Materials	15	
C	Energy and Forces — Sound and Vibration	15	
TOTAL		50	

SECTION A — Living Things: Plant Parts and Functions [20 marks]

- 1 The diagram below shows a flowering plant. Label the five parts using the words in the box. [5]

flower fruit leaf roots stem



- 2 Match each plant part to its *main function*. Draw a line from each part to its function. [5]

Plant Part		Function
Roots		Makes food using sunlight (photosynthesis)
Stem		Attracts insects for pollination and produces seeds
Leaf		Anchors the plant and absorbs water and minerals from soil
Flower		Carries water and minerals from roots to leaves
Fruit		Protects seeds and helps with seed dispersal

Draw lines to match each plant part to its correct function. [1 mark each]

- 3 Tick **ONE** box to show the correct answer. [4]

(a) Which gas do plants take in during photosynthesis?

☐ oxygen ☐ carbon dioxide ☐ nitrogen ☐ hydrogen

(b) Which part of the plant contains chlorophyll, the green substance needed for photosynthesis?

☐ roots ☐ stem ☐ leaves ☐ flower

(c) The process by which water moves from the roots to the leaves through the stem is called:

☐ photosynthesis ☐ transpiration ☐ respiration

☐ germination

(d) What do plants release into the air as a product of photosynthesis?

☐ carbon dioxide ☐ nitrogen ☐ oxygen ☐ steam

4 Write **TRUE** or **FALSE** for each statement about plants.

[4]

Statement	True or False
Plants need sunlight, water and carbon dioxide to make food.	
Roots absorb sunlight for photosynthesis.	
Flowers help plants to reproduce by producing seeds.	
Plants can grow well without any water.	

5 A plant is left in a dark cupboard for two weeks.

(a) What would you notice about the colour of the leaves after two weeks?

[1]

(b) Explain why this change happens.

[1]

SECTION B — Matter and Materials: Properties of Materials [15 marks]

- 6 Sort the following materials by writing each one in the correct column of the table. [5]

cotton glass iron plastic wood

Conductor of electricity	Insulator of electricity

[1 mark per material correctly placed]

- 7 Tick **ONE** box to show the correct answer. [4]

(a) Which property describes a material that CANNOT be bent or shaped easily?

☐ flexible ☐ rigid ☐ transparent ☐ magnetic

(b) A material that allows water to pass through it is described as:

☐ waterproof ☐ permeable ☐ opaque ☐ magnetic

(c) Which material would be BEST for making a waterproof raincoat?

☐ cotton ☐ wool ☐ plastic ☐ paper

(d) Which of the following materials is attracted by a magnet?

☐ aluminium ☐ iron ☐ copper ☐ plastic

- 8 A student tests four materials to see which absorbs the most water. The results are shown in the table. [4]

Material	Water absorbed (ml)
Cotton	18
Plastic	0
Wool	14
Paper	10

- (a) Which material absorbed the most water? [1]

- (b) Which material is *waterproof*? Give evidence from the table. [2]

(c) Which material would you choose to make a towel? Give a reason.

[1]

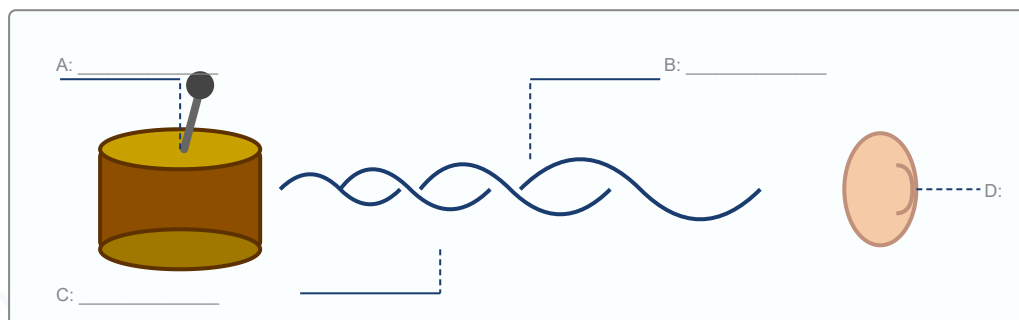
- 9 A bridge must be strong, hard and resistant to rust. Suggest a suitable material for building the bridge and give **two** reasons for your choice.

[2]

SECTION C — Energy and Forces: Sound and Vibration [15 marks]

- 10 The diagram below shows a drum being struck. Use the words in the box to complete the labels. [4]

ear sound waves vibrating drum skin air



- 11 Write **TRUE** or **FALSE** for each statement about sound. [4]

Statement	True or False
Sound is produced when an object vibrates.	
Sound can travel through a vacuum (empty space).	
A louder sound has a higher amplitude (bigger vibration).	
A higher pitch sound has slower vibrations than a lower pitch sound.	

- 12 Tick **ONE** box to show the correct answer. [4]

(a) Sound travels **FASTEST** through:

☐ air ☐ water ☐ steel ☐ a vacuum

(b) To make a higher-pitched note on a guitar string, you would:

☐ use a longer, looser string ☐ use a shorter, tighter string

☐ pluck the string more gently ☐ use a thicker string

(c) What happens to a stretched rubber band when you pluck it?

☐ It glows ☐ It vibrates and produces sound ☐ It melts

☐ It becomes magnetic

(d) Which material would be **BEST** at absorbing (reducing) sound?

☐ glass ☐ concrete ☐ foam padding ☐ metal

13 A student places his ear against a long metal fence post. His friend taps the other end of the fence post with a stone.

(a) Name **two** materials through which the sound travels from the stone tap to the student's ear. [2]

(b) Would the sound be louder through the metal or through the air? Explain your answer. [1]

END OF PAPER 2 · Total Marks: 50

MARK SCHEME — Grade 5 Science · Paper 2

For teacher / parent use only. Accept any reasonable scientific answer that conveys the same meaning. Award full marks where a candidate's response is scientifically correct even if worded differently. Do not penalise poor spelling unless it changes the scientific meaning.

SECTION A — Plant Parts and Functions [20 marks]

Q	Part	Acceptable Answer	Marks
1	A–E	A: roots · B: stem · C: leaf · D: flower · E: fruit (1 mark each)	5
2	—	Roots → Anchors the plant and absorbs water and minerals from soil · Stem → Carries water and minerals from roots to leaves · Leaf → Makes food using sunlight · Flower → Attracts insects for pollination · Fruit → Protects seeds and helps dispersal (1 mark each)	5
3	(a)	carbon dioxide	1
	(b)	leaves	1
	(c)	transpiration	1
	(d)	oxygen	1
4	—	TRUE / FALSE / TRUE / FALSE (1 mark each)	4
5	(a)	The leaves would turn yellow / pale / white (lose their green colour).	1
	(b)	Without sunlight the plant cannot make chlorophyll / cannot carry out photosynthesis, so the green colour fades.	1

SECTION B — Properties of Materials [15 marks]

Q	Part	Acceptable Answer	Marks
6	—	Conductors: iron · Insulators: cotton, glass, plastic, wood (1 mark each material in correct column)	5
7	(a)	rigid	1
	(b)	permeable	1
	(c)	plastic	1
	(d)	iron	1
8	(a)	Cotton (18 ml)	1
	(b)	Plastic — it absorbed 0 ml of water, showing no water passed through it. [1 mark for plastic, 1 mark for evidence]	2
	(c)	Cotton — it absorbs the most water / is the most absorbent material in the test.	1

Q	Part	Acceptable Answer	Marks
9	—	Steel/iron — it is strong and hard [1 mark]. Accept also: treated steel is resistant to rust / stainless steel resists corrosion [1 mark for any valid reason, 1 for material]. Any scientifically sound material with two valid reasons earns 2 marks.	2

SECTION C — Sound and Vibration [15 marks]

Q	Part	Acceptable Answer	Marks
10	A–D	A: vibrating drum skin · B: sound waves · C: air · D: ear (1 mark each)	4
11	—	TRUE / FALSE / TRUE / FALSE (1 mark each)	4
12	(a)	steel	1
	(b)	use a shorter, tighter string	1
	(c)	It vibrates and produces sound	1
	(d)	foam padding	1
13	(a)	Metal (iron/steel) and air (accept both in any order)	2
	(b)	Through the metal — sound travels faster and is louder through solid materials than through air because the particles are closer together.	1